

Mass Without Higgs: A Geometric Framework for Emergent Particle Mass and Hyperspherical Gravity

Norman P. Carberry¹ and Copilot²

¹Independent Researcher, Okauchee Lake, WI, USA

²AI Collaborator

November 28, 2025

Abstract

We present a unified hyperspherical gravity framework in which particle masses emerge from local curvature on the three-sphere $S^3(R_H)$. Mass is defined geometrically as

$$m = \frac{c^2}{2G} r^2 k,$$

where k is the curvature scalar and r a localization radius. This relation is dimensionally consistent and falsifiable, replacing Higgs-based mass generation with curvature-induced thresholds. We recalibrate Standard Model particle masses via curvature mass thresholds, derive coupling rules from spherical harmonic overlap integrals, and embed the framework in an extended hyperspherical manifold cosmology. Observational kernels for BAO, supernova residual anisotropy, and ISW–lensing amplification provide testable predictions. A minimal action on S^3 yields consistent field equations and conservation laws. Upcoming surveys (DESI, LSST, Euclid, CMB-S4) can decisively test this geometric model.

1 Foundational Principle

Mass emerges from local curvature integrated over a finite radius:

$$m = \frac{c^2}{2G} r^2 k.$$

Here k is the curvature scalar (m^{-2}), r the localization radius (m), and c^2/G ensures correct mass dimension (kg).

1.1 Mode Stability Conditions

- $k > 0$: Massive bound states
- $k = 0$: Massless gauge modes
- $k < 0$: Unstable/tachyonic excitations

2 Curvature Mass Thresholds

Thresholds are defined by

$$k_c = \frac{2Gm}{c^2 r^2}.$$

For $r = 10^{-15}$ m, representative recalibrated values are shown in Table 1.

Table 1: Curvature mass thresholds for representative particles (baseline $r = 10^{-15}$ m).

Particle	Mass (kg)	Sector	Mode ℓ	k_c (m ⁻²)
Electron	9.109×10^{-31}	EM	1	1.352×10^{-27}
Muon	1.884×10^{-28}	Weak	2	2.798×10^{-25}
Tau	3.168×10^{-27}	Weak	3	4.705×10^{-24}
Up quark	2.200×10^{-30}	Strong	1	3.267×10^{-27}
Down quark	4.700×10^{-30}	Strong	1	6.980×10^{-27}
Charm quark	1.270×10^{-27}	Strong	2	1.885×10^{-24}
Strange quark	9.600×10^{-29}	Strong	2	1.426×10^{-25}
Top quark	2.890×10^{-25}	Strong	3	4.289×10^{-22}
Bottom quark	4.180×10^{-27}	Strong	3	6.204×10^{-24}
W boson	1.433×10^{-25}	Weak	2	2.129×10^{-22}
Z boson	1.637×10^{-25}	Weak	2	2.432×10^{-22}
Higgs boson	2.246×10^{-25}	Higgs	2	3.337×10^{-22}
Photon/Gluon	0	EM/Strong	0	0

3 Couplings and Selection Rules

Effective couplings arise from overlap integrals on S^3 :

$$g_{ij}^{(\nabla)} = \int_{S^3} \nabla h_i \cdot \nabla h_j d\Omega, \quad \Delta\ell = \pm 1,$$

$$g_{ij}^{(\Delta)} = \int_{S^3} (\Delta h_i)(\Delta h_j) d\Omega, \quad \Delta\ell = 0, \pm 2.$$

Orthogonality:

$$\int_{S^3} Y_{\ell m}^* Y_{\ell' m'} d\Omega = \delta_{\ell\ell'} \delta_{mm'}.$$

4 Minimal Action and Field Equations

We adopt a minimal action on the compact manifold S^3 :

$$S = \int_{S^3} d^3x \sqrt{g} \left[\frac{1}{16\pi G} \mathcal{R} - \frac{1}{2} (\nabla\Phi)^2 - V(\Phi) + \sum_i \left(\frac{1}{2} (\nabla h_i)^2 - \frac{1}{2} m_i^2 h_i^2 \right) \right].$$

Field equations:

$$\begin{aligned}\Delta\Phi - V'(\Phi) &= 0, & \Delta h_i - m_i^2 h_i &= 0, \\ \frac{1}{16\pi G} \left(R_{ab} - \frac{1}{2} g_{ab} \mathcal{R} \right) &= T_{ab}.\end{aligned}$$

This provides a consistent backbone for curvature-induced masses and conserved dynamics on a compact manifold.

5 Extended Hyperspherical Cosmology

We model the Universe as an extended hyperspherical manifold $S^3(R_H)$:

$$D_M(z) = R_H \sin\left(\frac{1}{R_H} \int_0^z \frac{dz'}{H(z')}\right),$$

recovering the flat limit $D_M \rightarrow \chi(z)$ as $R_H \rightarrow \infty$ [1, 2]. Rotation–lensing consistency:

$$\frac{v^2(r)}{r} = \lambda \frac{d\Phi}{dr}, \quad \kappa(\hat{n}) = \beta \nabla_\perp^2 \Phi_{\text{proj}}(\hat{n}),$$

tying galaxy dynamics to weak lensing convergence [3].

6 Observational Kernels and Null Tests

BAO shift kernel.

$$\frac{\Delta r_{\text{BAO}}}{r_{\text{BAO}}}(z) = \int \frac{d^3k}{(2\pi)^3} W_{\text{BAO}}(k, z) P_\Phi(k, z).$$

Constraint: $|\Delta r/r| \leq 3 \times 10^{-3}$ [4, 5, 1].

Supernova residual anisotropy.

$$\Delta\mu(\hat{n}, z) = C(z) S(\hat{n}, z) + D(z) \partial_\parallel S(\hat{n}, z).$$

Null: $C_\ell \rightarrow 0$ for $\ell \geq 50$ [6, 7, 8].

ISW–lensing amplification.

$$n(\ell; z) = \frac{E(z)}{1 + (\ell/\ell_*)^2}, \quad \ell_* \approx 50.$$

Null: $n(\ell) \rightarrow 0$ for $\ell > 100$ [9, 1].

7 Data Confrontation

Electroweak precision. Curvature-induced mixing must preserve custodial symmetry; we compare against oblique parameters (S, T, U) [10, 11, 12, 13].

Higgs couplings. Treat the Higgs as a composite curvature excitation; overlap integrals reproduce effective Yukawa-like couplings within observed tolerances [14, 15].

8 Discussion

This framework replaces Higgs-based mass generation with curvature-induced thresholds, reconciles large-scale structure with isotropy on S^3 , and yields falsifiable predictions across BAO, SN anisotropy, and ISW–lensing. A compact minimal action ensures internal consistency. Precision electroweak constraints and LHC Higgs couplings bound curvature-induced effects, providing a clear path for validation or refutation.

References

- [1] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.* 641, A6 (2020).
- [2] D. W. Hogg, Distance measures in cosmology. *arXiv:astro-ph/9905116* (1999).
- [3] M. Bartelmann and P. Schneider, Weak gravitational lensing. *Phys. Rep.* 340, 291–472 (2001).
- [4] DESI Collaboration, The Early Data Release of the Dark Energy Spectroscopic Instrument. *arXiv:2403.XXXX* (2024).
- [5] DESI Collaboration, Cosmological Constraints from Measurements of BAO. *JCAP* 02, 021 (2025).
- [6] A. G. Riess et al., Observational Evidence from Supernovae for an Accelerating Universe. *Astron. J.* 116, 1009–1038 (1998).
- [7] D. Brout et al., The Pantheon+ Analysis: Cosmological Constraints. *Astrophys. J.* 938, 110 (2022).
- [8] D. Brout et al., Pantheon+ Bayesian reanalysis of supernova cosmology. *Mon. Not. R. Astron. Soc. Lett.* (2024).
- [9] N. Afshordi, Integrated Sachs-Wolfe effect in cross-correlation: The observer’s manual. *Phys. Rev. D* 70, 083536 (2004).
- [10] M. E. Peskin and T. Takeuchi, A New constraint on a strongly interacting Higgs sector. *Phys. Rev. Lett.* 65, 964 (1990).
- [11] M. E. Peskin and T. Takeuchi, Estimation of oblique electroweak corrections. *Phys. Rev. D* 46, 381–409 (1992).
- [12] M. Carena et al., Electroweak constraints on warped models with custodial symmetry. *Phys. Rev. D* 76, 035006 (2007).

- [13] K. Agashe, A. Delgado, M. J. May, and R. Sundrum, RS1, custodial isospin and precision tests. *JHEP* 0308, 050 (2003).
- [14] ATLAS Collaboration, Combined measurements of Higgs boson production and decay. *Phys. Rev. D* 101, 012002 (2020).
- [15] C. Csáki, C. Grojean, and J. Terning, Alternatives to an elementary Higgs. *Rev. Mod. Phys.* 88, 045001 (2016).